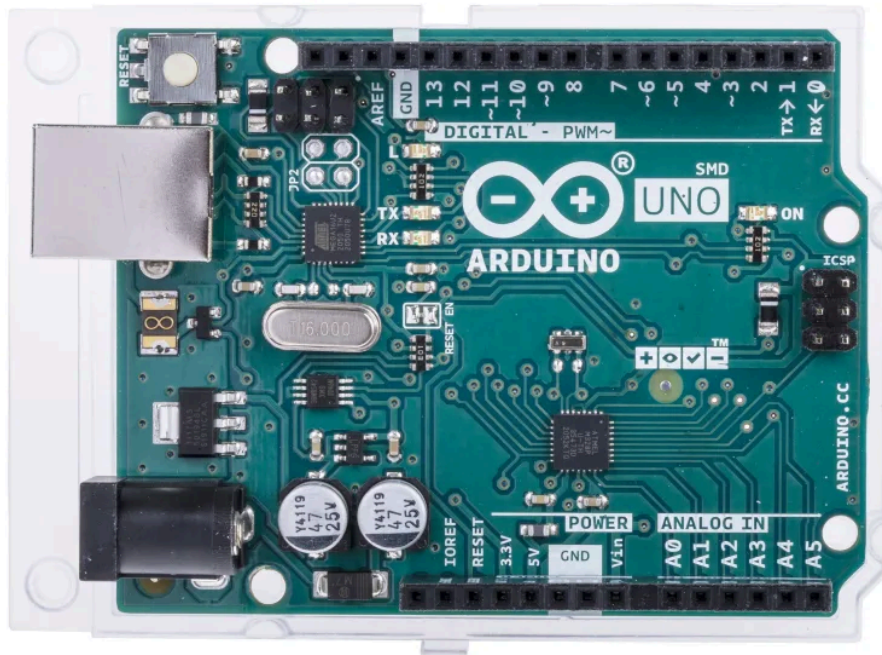


Arduino UNO R3 - Motor Code:



```
// Pin definitions for L293D
const int ENA = 9;    // PWM pin for speed
const int IN1 = 8;    // Direction 1
const int IN2 = 7;    // Direction 2

int rotateTime = 1200; // milliseconds
int motorSpeed = 150;  // 0-255

void setup() {
  pinMode(ENA, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);

  // Stop motor at start
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);
  analogWrite(ENA, 0);
}

void loop() {

  // ----- Spin Forward -----
```

```

digitalWrite(IN1, HIGH);
digitalWrite(IN2, LOW);
analogWrite(ENA, motorSpeed);
delay(rotateTime);

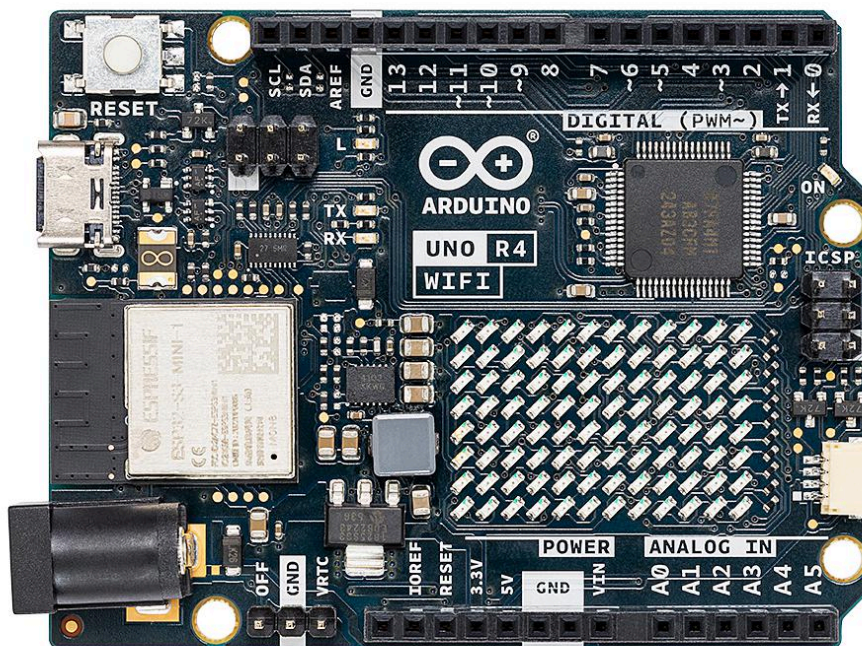
// Stop briefly
analogWrite(ENA, 0);
delay(200);

// ----- Spin Backward -----
digitalWrite(IN1, LOW);
digitalWrite(IN2, HIGH);
analogWrite(ENA, motorSpeed);
delay(rotateTime); // same time in opposite direction

// Stop briefly
analogWrite(ENA, 0);
delay(200);
}

```

Arduino UNO R4 Wifi - App Code:



```
#include <WiFiS3.h>
#include <ArduinoGraphics.h>
#include <Arduino_LED_Matrix.h>
#include <LiquidCrystal.h>

ArduinoLEDMatrix matrix;

// AP credentials
char ssid[] = "BusStop_AP";
char pass[] = "busstop123";

WiFiServer server(80);

bool stopRequested = false;
String currentBus = "";
String scrollText = " Waiting for bus... ";

// timing
unsigned long lastMatrixScrollMillis = 0;
unsigned long lastLcdScrollMillis = 0;

const unsigned long lcdScrollInterval = 15;
const unsigned long matrixScrollInterval = 60;

// Pin mapping LCD: RS, E, D4, D5, D6, D7
LiquidCrystal lcd(2, 3, 4, 5, 6, 7);

//button pin
const int buttonPin = 12; // button between D12 and GND
const int ledPin = 13; // LED + resistor on D13, other leg to GND
bool ledActive = false;
unsigned long ledEndTime = 0;
int lastButtonState = HIGH; // using INPUT_PULLUP, so HIGH = not
pressed

void setup() {
    Serial.begin(115200);
```

```

pinMode(ledPin, OUTPUT);
digitalWrite(ledPin, LOW);

// Configure button with internal pull-up
pinMode(buttonPin, INPUT_PULLUP);

// Access Point Network Config
Serial.println("Starting Access Point...");
int status = WiFi.beginAP(ssid, pass);
Serial.print("AP status: ");
Serial.println(status);
Serial.print("AP IP: ");
Serial.println(WiFi.localIP());

server.begin();

// LED matrix init
matrix.begin();
matrix.textFont(Font_5x7);
matrix.textScrollSpeed(60); // leave matrix speed as-is

// LCD init
lcd.begin(16, 2); // 16x2 LCD
lcd.clear();
lcd.setCursor(0, 0);
lcd.print(scrollText); // print initial message once
}

void loop() {
  //handles Phone App
  handleClient();
  //Handles display
  updateScrollingText();
  //button control
  int buttonState = digitalRead(buttonPin);

  if (lastButtonState == HIGH && buttonState == LOW) {
    // Button just pressed
    // Turn LED on
    digitalWrite(ledPin, HIGH);
    ledActive = true;
  }
}

```

```

        // Set timer for 10 seconds from now
        ledEndTime = millis() + 10000UL;

        // Update LCD
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("Call Light ON");
    }
    lastButtonState = buttonState;
    // If LED is active and time has expired
    if (ledActive && (long)(millis() - ledEndTime) >= 0) {
        digitalWrite(ledPin, LOW);
        ledActive = false;
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print(scrollText);
    }
}

// ----- HTTP handling -----
void handleClient() {
    WiFiClient client = server.available();
    if (!client) return;

    // Don't block forever waiting for data:
    unsigned long startWait = millis();
    while (!client.available() && (millis() - startWait < 200)) {
        // keep scrolling while we wait a bit for data
        updateScrollingText();
        delay(1);
    }

    if (!client.available()) {
        // no data in time, bail out so scrolling stays smooth
        return;
    }

    String req = client.readStringUntil('\r');
    client.read(); // '\n'
    Serial.print("Request: ");
    Serial.println(req);

```

```
bool isStop = req.startsWith("POST /stop");
bool isClear = req.startsWith("POST /clear");

if (isStop) {
    String bus = extractBusFromRequest(req);
    if (bus.length() > 0) {
        currentBus = bus;
        stopRequested = true;

        scrollText = " Stopping bus " + currentBus + " next ";

        // refresh LCD with new text
        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print(scrollText);

        Serial.print("Selected bus: ");
        Serial.println(currentBus);
    }
}

if (isClear) {
    stopRequested = false;
    currentBus = "";
    scrollText = " Waiting for bus... ";

    // refresh LCD with new text
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print(scrollText);
}

// discard the rest of headers
while (client.available()) {
    String line = client.readStringUntil('\n');
    if (line == "\r" || line.length() == 1) break;
}

// Simple HTML response
client.println("HTTP/1.1 200 OK");
client.println("Content-Type: text/html");
client.println("Connection: close");
client.println();
```

[illegible]

```

        lcd.scrollDisplayLeft(); // Shift LCD left by 1 position
        lcd.scrollDisplayLeft(); // Shift LCD left by 1 position
        lcd.scrollDisplayLeft(); // Shift LCD left by 1 position
        lcd.scrollDisplayLeft(); // Shift LCD left by 1 position
    }

    // ---- LED MATRIX SCROLLING (unchanged speed) ----
    if (now - lastMatrixScrollMillis < matrixScrollInterval) {
        return; // wait until it's time to move matrix again
    }

    lastMatrixScrollMillis = now;

    matrix.beginDraw();
    matrix.stroke(0xFFFFFFFF); // white
    matrix.beginText(matrix.width(), 1, 0xFFFFFF);
    matrix.print(scrollText);
    matrix.endText(SCROLL_LEFT);
    matrix.endDraw();
}

```

App:

MainActivity_kt – Backend:

```

package com.example.busstopcaller
import android.os.Bundle
import android.view.View
import android.widget.*
import androidx.appcompat.app.AppCompatActivity
import java.net.HttpURLConnection
import java.net.URL
import kotlin.random.Random

data class Bus(val number: String, var eta: Int)

class MainActivity : AppCompatActivity() {

    private val baseUrl = "http://192.168.4.1"

    private lateinit var busListView: ListView
    private lateinit var statusText: TextView
    private lateinit var btnClearStop: Button
    private lateinit var searchView: SearchView
    private lateinit var searchCard: View
    private lateinit var btnSearchBus: Button
    private lateinit var departuresHeader: View

```



```

private lateinit var listCard: View
private lateinit var mapView: ImageView

private lateinit var tabSearch: TextView
private lateinit var tabDepartures: TextView
private lateinit var tabMaps: TextView
private lateinit var underlineSearch: View
private lateinit var underlineDepartures: View
private lateinit var underlineMaps: View

private var allBuses = mutableListOf<Bus>()
private var filteredBuses = mutableListOf<Bus>()
private lateinit var adapter: ArrayAdapter<String>

enum class Tab { SEARCH, DEPARTURES, MAPS }

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContentView(R.layout.activity_main)

    // Views
    busListView = findViewById(R.id.busListView)
    statusText = findViewById(R.id.statusText)
    btnClearStop = findViewById(R.id.btnClearStop)
    searchView = findViewById(R.id.searchView)
    searchCard = findViewById(R.id.searchCard)
    btnSearchBus = findViewById(R.id.btnSearchBus)
    departuresHeader = findViewById(R.id.departuresHeader)
    listCard = findViewById(R.id.listCard)
    mapView = findViewById(R.id.mapView)

    tabSearch = findViewById(R.id.tabSearch)
    tabDepartures = findViewById(R.id.tabDepartures)
    tabMaps = findViewById(R.id.tabMaps)
    underlineSearch = findViewById(R.id.underlineSearch)
    underlineDepartures = findViewById(R.id.underlineDepartures)
    underlineMaps = findViewById(R.id.underlineMaps)

    // Create bus list with random ETAs
    allBuses = mutableListOf(
        Bus("4", randEta()),
        Bus("7", randEta()),
        Bus("9", randEta()),
        Bus("11", randEta()),
        Bus("13", randEta()),
        Bus("16", randEta()),
        Bus("27", randEta()),
        Bus("39A", randEta()),
    )
}

```

```

        Bus("46A", randEta()),
        Bus("145", randEta())
    )

    filteredBuses.addAll(allBuses)

    adapter = object : ArrayAdapter<String>(
        this,
        R.layout.item_bus,
        R.id.busNumberText,
        filteredBuses.map { "${it.number}    (${it.eta} min)" }
    ) { }

    busListView.adapter = adapter

    busListView.setOnItemClickListener { _, _, position, _ ->
        val bus = filteredBuses[position]
        requestStopForBus(bus)
    }

    btnClearStop.setOnClickListener {
        sendCommand("/clear")
    }

    // Optional: "Search bus" button just re-applies filter and
maybe reshuffles ETAs
    btnSearchBus.setOnClickListener {
        // Regenerate ETAs to simulate new departure board
        allBuses.forEach { it.eta = randEta() }
        filterList(searchView.query?.toString() ?: "")
    }

    // Search filtering as you type
    searchView.setOnQueryTextListener(object :
SearchView.OnQueryTextListener {
        override fun onQueryTextChange(newText: String?):
Boolean {
            filterList(newText ?: "")
            return true
        }

        override fun onQueryTextSubmit(query: String?): Boolean
= false
    })

    // Tab clicks
    tabSearch.setOnClickListener { setActiveTab(Tab.SEARCH) }

```

```

        tabDepartures.setOnClickListener {
setActiveTab(Tab.DEPARTURES) }
        tabMaps.setOnClickListener { setActiveTab(Tab.MAPS) }

        // Start in Search tab
        setActiveTab(Tab.SEARCH)
    }

    private fun randEta(): Int = Random.nextInt(1, 21)

    private fun filterList(query: String) {
        filteredBuses.clear()
        filteredBuses.addAll(
            allBuses.filter { it.number.contains(query, ignoreCase =
true) }
        )
        adapter.clear()
        adapter.addAll(filteredBuses.map { "${it.number}    (${it.eta}
min)" })
        adapter.notifyDataSetChanged()
    }

    private fun requestStopForBus(bus: Bus) {
        statusText.text = "Requesting stop for bus
${bus.number}..."
        sendCommand("/stop?bus=${bus.number}&eta=${bus.eta}")
    }

    private fun sendCommand(path: String) {
        Thread {
            try {
                val url = URL("$baseUrl$path")
                val conn = (url.openConnection() as
HttpURLConnection).apply {
                    requestMethod = "POST"
                    doOutput = true
                    connectTimeout = 5000
                    readTimeout = 5000
                }

                conn.outputStream.use { }

                val code = conn.responseCode
                val body = conn.inputStream.bufferedReader().use {
it.readText() }

                runOnUiThread {
                    statusText.text = "HTTP $code\n$body"

```

```

        }

        conn.disconnect()
    } catch (e: Exception) {
        runOnUiThread {
            textStatus.text = "Error: ${e.message}"
        }
    }
}.start()
}

private fun setActiveTab(tab: Tab) {
    val primary = getColor(R.color.tfi_text_primary)
    val secondary = getColor(R.color.tfi_text_secondary)

    tabSearch.setTextColor(if (tab == Tab.SEARCH) primary else
secondary)
    tabDepartures.setTextColor(if (tab == Tab.DEPARTURES)
primary else secondary)
    tabMaps.setTextColor(if (tab == Tab.MAPS) primary else
secondary)

    underlineSearch.visibility = if (tab == Tab.SEARCH)
View.VISIBLE else View.INVISIBLE
    underlineDepartures.visibility = if (tab == Tab.DEPARTURES)
View.VISIBLE else View.INVISIBLE
    underlineMaps.visibility = if (tab == Tab.MAPS) View.VISIBLE
else View.INVISIBLE

    when (tab) {
        Tab.SEARCH -> {
            searchCard.visibility = View.VISIBLE
            btnSearchBus.visibility = View.VISIBLE
            departuresHeader.visibility = View.VISIBLE
            listCard.visibility = View.VISIBLE
            mapView.visibility = View.GONE
            btnClearStop.visibility = View.VISIBLE
        }
        Tab.DEPARTURES -> {
            searchCard.visibility = View.GONE
            btnSearchBus.visibility = View.GONE
            departuresHeader.visibility = View.VISIBLE
            listCard.visibility = View.VISIBLE
            mapView.visibility = View.GONE
            btnClearStop.visibility = View.VISIBLE
        }
        Tab.MAPS -> {
            searchCard.visibility = View.GONE

```

```

        btnSearchBus.visibility = View.GONE
        departuresHeader.visibility = View.GONE
        listCard.visibility = View.GONE
        mapView.visibility = View.VISIBLE
        btnClearStop.visibility = View.GONE
    }
}
}
}
}

```

Item_bus.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="64dp"
    android:orientation="horizontal"
    android:gravity="center_vertical"
    android:paddingStart="12dp"
    android:paddingEnd="12dp"
    android:layout_margin="4dp"
    android:background="@color/tfi_bg_dark">

    <!-- Left colored strip (TFI green) -->
    <View
        android:layout_width="4dp"
        android:layout_height="match_parent"
        android:background="@color/tfi_green"
        android:layout_marginEnd="8dp" />

    <!-- Content -->
    <LinearLayout
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:orientation="vertical">

        <TextView
            android:id="@+id/busNumberText"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="13"
            android:textSize="18sp"
            android:textStyle="bold"
            android:textColor="@color/tfi_text_primary" />

        <TextView
            android:id="@+id/busEtaText"

```

```

        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="7 min"
        android:textSize="13sp"
        android:textColor="@color/tfi_text_secondary" />
    </LinearLayout>

    <!-- star icon area (placeholder) -->
    <TextView
        android:id="@+id/busFavStar"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="★"
        android:textColor="@color/tfi_yellow"
        android:textSize="16sp"
        android:paddingStart="8dp"
        android:paddingEnd="4dp"
        android:visibility="gone" />
</LinearLayout>

```

colors.xml:

```

<resources>
    <!-- TFI brand-ish colors -->
    <color name="tfi_green">#00A651</color>        <!-- main accent -->
    <color name="tfi_yellow">#FFC20A</color>
    <color name="tfi_blue">#0070B8</color>

    <!-- Dark theme base -->
    <color name="tfi_bg_dark">#101418</color>        <!-- background
-->
    <color name="tfi_card_dark">#181D23</color>        <!-- cards /
panels -->
    <color name="tfi_border">#262C34</color>
    <color name="tfi_text_primary">#FFFFFF</color>
    <color name="tfi_text_secondary">#B0BEC5</color>

    <!--defaults-->
    <color name="black">#000000</color>
    <color name="white">#FFFFFF</color>
</resources>

```

AndroidManifest.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<manifest
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission android:name="android.permission.INTERNET" />
    <application

```

```

        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.BusStopCaller"
        android:usesCleartextTraffic="true">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"
            />

                <category
android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

</manifest>

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@color/tfi_bg_dark">

    <!-- Logo in the top-right -->
    <ImageView
        android:id="@+id/logoImage"
        android:layout_width="90dp"
        android:layout_height="wrap_content"
        android:adjustViewBounds="true"
        android:src="@drawable/tfi_logo"
        android:contentDescription="@string/app_name"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="16dp"
        android:layout_marginEnd="16dp" />

    <!-- Header greeting -->
    <TextView

```

```
    android:id="@+id/headerGreeting"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Good day"
    android:textColor="@color/tfi_text_primary"
    android:textSize="22sp"
    android:textStyle="bold"
    android:paddingStart="20dp"
    android:paddingTop="20dp"
    android:paddingBottom="4dp"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintStart_toStartOf="parent" />
```

```
<!-- Small app label -->
```

```
<TextView
    android:id="@+id/headerSub"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="TFI Bus Stop"
    android:textColor="@color/tfi_text_secondary"
    android:textSize="14sp"
    android:paddingStart="20dp"
    android:paddingBottom="12dp"
    app:layout_constraintTop_toBottomOf="@id/headerGreeting"
    app:layout_constraintStart_toStartOf="parent" />
```

```
<!-- Tab row -->
```

```
<LinearLayout
    android:id="@+id/tabRow"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:paddingStart="20dp"
    android:paddingEnd="20dp"
    app:layout_constraintTop_toBottomOf="@id/headerSub"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent">
```

```
    <TextView
        android:id="@+id/tabSearch"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Search"
        android:textSize="14sp"
        android:textColor="@color/tfi_text_primary"
        android:paddingEnd="24dp" />
```

```
<TextView
```



```

        android:id="@+id/tabDepartures"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Departures"
        android:textSize="14sp"
        android:textColor="@color/tfi_text_secondary"
        android:paddingEnd="24dp" />

<TextView
    android:id="@+id/tabMaps"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Maps"
    android:textSize="14sp"
    android:textColor="@color/tfi_text_secondary" />
</LinearLayout>

<!-- Underlines for active tab -->
<View
    android:id="@+id/underlineSearch"
    android:layout_width="0dp"
    android:layout_height="2dp"
    android:background="@color/tfi_green"
    app:layout_constraintTop_toBottomOf="@id/tabRow"
    app:layout_constraintStart_toStartOf="@id/tabSearch"
    app:layout_constraintEnd_toEndOf="@id/tabSearch" />

<View
    android:id="@+id/underlineDepartures"
    android:layout_width="0dp"
    android:layout_height="2dp"
    android:background="@color/tfi_green"
    android:visibility="invisible"
    app:layout_constraintTop_toBottomOf="@id/tabRow"
    app:layout_constraintStart_toStartOf="@id/tabDepartures"
    app:layout_constraintEnd_toEndOf="@id/tabDepartures" />

<View
    android:id="@+id/underlineMaps"
    android:layout_width="0dp"
    android:layout_height="2dp"
    android:background="@color/tfi_green"
    android:visibility="invisible"
    app:layout_constraintTop_toBottomOf="@id/tabRow"
    app:layout_constraintStart_toStartOf="@id/tabMaps"
    app:layout_constraintEnd_toEndOf="@id/tabMaps" />

<!-- Search card (only visible in Search tab) -->

```

```

<LinearLayout
    android:id="@+id/searchCard"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:background="@color/tfi_card_dark"
    android:padding="12dp"
    android:layout_marginStart="20dp"
    android:layout_marginEnd="20dp"
    android:layout_marginTop="16dp"
    android:elevation="4dp"
    app:layout_constraintTop_toBottomOf="@id/underlineSearch"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Bus search"
        android:textSize="14sp"
        android:textColor="@color/tfi_text_secondary"
        android:paddingBottom="4dp" />

    <SearchView
        android:id="@+id/searchView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:iconifiedByDefault="false"
        android:queryHint="Search bus number"
        android:background="@color/tfi_card_dark" />

</LinearLayout>

<!-- (Optional) search button, mostly visual -->
<Button
    android:id="@+id/btnSearchBus"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:text="Search bus"
    android:textAllCaps="false"
    android:textSize="16sp"
    android:textColor="@color/tfi_text_primary"
    android:backgroundTint="@color/tfi_blue"
    android:layout_marginStart="20dp"
    android:layout_marginEnd="20dp"
    android:layout_marginTop="12dp"
    app:layout_constraintTop_toBottomOf="@id/searchCard"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent" />

```

```

<!-- Departures header -->
<TextView
    android:id="@+id/departuresHeader"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Departures"
    android:textColor="@color/tfi_text_secondary"
    android:textSize="14sp"
    android:layout_marginStart="20dp"
    android:layout_marginTop="20dp"
    app:layout_constraintTop_toBottomOf="@id/btnSearchBus"
    app:layout_constraintStart_toStartOf="parent" />

<!-- List container (search + departures modes) -->
<LinearLayout
    android:id="@+id/listCard"
    android:layout_width="0dp"
    android:layout_height="0dp"
    android:orientation="vertical"
    android:background="@color/tfi_card_dark"
    android:layout_marginStart="20dp"
    android:layout_marginEnd="20dp"
    android:layout_marginTop="8dp"
    android:layout_marginBottom="60dp"
    app:layout_constraintTop_toBottomOf="@id/departuresHeader"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintBottom_toTopOf="@id/statusText">

    <ListView
        android:id="@+id/busListView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:divider="@color/tfi_border"
        android:dividerHeight="1dp"
        android:padding="4dp" />
</LinearLayout>

<!-- Dublin map (only visible in Maps tab) -->
<ImageView
    android:id="@+id/mapView"
    android:layout_width="0dp"
    android:layout_height="0dp"
    android:src="@drawable/map_dublin"
    android:scaleType="centerCrop"
    android:visibility="gone"
    android:layout_marginStart="20dp"

```

```

        android:layout_marginEnd="20dp"
        android:layout_marginTop="16dp"
        android:layout_marginBottom="60dp"
        app:layout_constraintTop_toBottomOf="@id/underlineSearch"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintBottom_toTopOf="@id/statusText" />

<!-- Clear button, used in Search + Departures tabs -->
<Button
    android:id="@+id/btnClearStop"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Clear stop"
    android:textAllCaps="false"
    android:textColor="@color/tfi_text_primary"
    android:backgroundTint="@color/tfi_card_dark"
    android:layout_marginEnd="16dp"
    app:layout_constraintBottom_toTopOf="@id/statusText"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginBottom="4dp" />

<!-- Status footer -->
<TextView
    android:id="@+id/statusText"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:text="Status: idle"
    android:textColor="@color/tfi_text_secondary"
    android:textSize="13sp"
    android:padding="10dp"
    android:background="@color/tfi_bg_dark"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>

```

Github: <https://github.com/Guru-Chappalle/BusStopCaller>